

Technical Manual: CCR Traffic Light System

Versions:

- 3-Light (Red, Yellow, Green)
 - 2-Light (Red, Green)
-

1. System Overview

This system consists of modular traffic light units powered by a **Seeed Studio Xiao** microcontroller and controlled WS2812B addressable LEDs. Housed entirely in **3D printed enclosures**, it is designed for safe, educational, or demonstration purposes as part of the **CCR (Connected Crosswalk Response)** project.

2. Electrical Architecture

2.1 LED Structure

- Each light (Red, Yellow, Green) contains **3 WS2812B LEDs**, cut from an LED strip.
- LEDs are connected **in series** and driven by individual microcontroller pins.

2.2 Pin Mapping (3-Light)

cpp

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```
#define RED_PIN      D8
#define YELLOW_PIN   D4
#define GREEN_PIN    D5
```

In the 2-light version, only D8 (Red) and D5 (Green) are used.

2.3 Controller

- **Microcontroller:** Seeed Studio Xiao (ARM Cortex-M0+)
- **Power:** Supplied via **USB Type-C panel mount connector** on the back

No resistors or capacitors are used in this system.

3. Mechanical Assembly

3.1 Light Unit Housing

Each color light includes:

- **3D Printed Light Body** with one end closed (top or bottom)
- **3D Printed Diffuser Cap** on the front
- WS2812B LED strip (3 LEDs) mounted inside
- Internal wire channel for clean routing

3.2 Traffic Light Body

Stacked structure based on number of lights:

Segment	Design Description
Top Segment	Closed bottom 3D printed body
Middle Unit	Open on both ends (acts as joiner) — 3-light only
Bottom Unit	Closed top 3D printed body

Each unit is stacked vertically and aligned with internal slots or guides.

3.3 Rear Panel Configuration

Rear Panel	Purpose
Panel A	Mounts the USB Type-C connector (RS 2188312)
Panel B	3D printed mounting plate with screw holes for wall/plate attachment

Both panels are 3D printed and fixed to the back of the light stack.

4. Bill of Materials (BOM)

#	Component	3-Light Qty	2-Light Qty	Description
1	Seed Studio Xiao	1	1	Main controller

2	WS2812B LED Strip	9 LEDs	6 LEDs	Cut in 3s
3	USB Type-C Panel Mount (RS 2188312)	1	1	Mounted on back
4	3D Printed Light Bodies (Closed)	2	2	Top & bottom housings
5	3D Printed Joiner (Open both ends)	1	0	For middle stacking (3-light only)
6	3D Printed Diffuser Caps	3	2	One per light unit
7	3D Printed Back Panel – Type C Mount	1	1	Holds USB Type-C
8	3D Printed Mounting Plate (Panel B)	1	1	For surface or wall mounting
9	Jumper Wires / Silicone Wires	As needed	As needed	For LED and power connections
10	Screws / Glue	As needed	As needed	Assembly aids

5. Notes

- Ensure wiring polarity (5V, GND, Data) is respected for each LED group.
- Avoid overloading Xiao's onboard 5V line — keep brightness reasonable or power externally if needed.
- 3D printed components should be printed with **high infill (~50%)** for mechanical strength, especially mounting parts.